NATIONAL CYBERWATCH

MID-ATLANTIC CCDC 2013

WWW.MACCDC.ORG

TEAM PACKET v1.0

TABLE OF CONTENTS

Competition Schedule 2
CCDC Mission 3
Competition Objectives 3
Terminology 3
Competition Rules 4
- Competitor Eligibility 4
- Team Composition 4
- Team Representatives 5
- Competition Conduct 5
- Internet Usage 6
- Permitted Materials 7
- Professional Conduct 7
- Questions, Disputes, and Disclosures 8
- Red Team Attack Rules 8

Business Scenario 9
E-Voting Scenario 9
Red Team Goals 9
Scoring 10
- General Information 10
- Tie Breaker 11
- Service Rounds 11
- Dependent Services 11
- Non-Service Scored Assets 12
- Service Data Confidentiality 12
- Red Team 12
- Injects 12
- Supervisor Reporting 13
- Incident Reporting 13

Defender Network 14
IP Address Table 14
System Information 15
- Firewall 15
- DMZ Segment 15
- Outside Segment 16
- Hidden Assets 16

INFORMATION CONTAINED IN THIS DOCUMENT IS SUBJECT TO CHANGE

COMPETITION SCHEDULE

Wednesday, April 10	
9:00 AM - 3:15 PM	Speaker Symposium – "*Securing the Vote: Electronic Voting Systems and Cybersecurity*" - Auditorium
3:15 PM - 5:00 PM	Speaker Symposium Networking Reception (Off-Site at La Palapa Too)
Thursday, April 11	
8:30 AM - 10:00 AM	Blue Team Check-In Open
10:00 AM - 12:00 PM	Introductions and Team Briefings - Auditorium
	Blue Team competition area open to teams
12:00 PM - 12:30 PM	Lunch Break (On Your Own)
12:30 PM - 1:30 PM	Sponsor Briefings - Auditorium
1:30 PM - 3:30 PM	Job Fair – Mezzanine Level
Friday, April 12	
7:00 AM	Volunteer Check-In/Briefing
7:30 AM - 8:30 AM	Blue Team & Red Team Check-In Open
8:30 AM - 9:00 AM	Daily Briefing – Auditorium
9:00 AM - 5:00 PM	Day 1 MACCDC Competition
12:00 PM - 1:00 PM	Lunch (on your own)
12:00 PM - 12:30 PM	Guest Speaker: Ron Gula, CEO Tenable Network Security: "*Careers and Opportunities in Cybersecurity*" - Auditorium
12:30 PM - 1:00 PM	Guest Speaker: TBD – Auditorium
5:00 PM	Day one competition ends
5:00 PM - 5:30 PM	Daily Debrief – Auditorium
Saturday, April 13	
7:00 AM	Volunteer Check-In/Briefing
7:00 AM - 8:00 AM	Team Check-In
8:30 AM - 9:00 AM	Daily Briefing - Auditorium
9:00 AM - 5:00 PM	Day 2 MACCDC Competition
9:00 AM - 12:00 PM	High School Cybersecurity Fair and Expo
12:00 PM - 1:00 PM	Lunch (on your own)
12:00 PM - 12:30 PM	Guest Speaker: TBD - Auditorium
12:30 PM - 1:00 PM	Guest Speaker: TBD - Auditorium
5:00 PM	Day Two Competition Ends
6:30 PM - 8:00 PM	Competition Debrief and Awards Ceremony

CCDC MISSION

"The goal of a Cyber Defense Competition is to provide hands-on application of information assurance skills; as such, they enhance students' understanding of both theory and practice. They provide students a laboratory in which to experiment, just as in other fields of science. They fulfill the same role as capstone projects in a traditional engineering program (i.e., projects that allow students to synthesize and integrate knowledge acquired through course work and other learning experiences into a project usually conducted in a workplace). The competition combines legal, ethical, forensics, and technical components while emphasizing a team approach. Such experiential education increases the knowledge and expertise of future professionals who may be in a position to contribute to the secure design and operation of critical information and its supporting infrastructure" (from *Exploring a National Cyber Security Exercise for Colleges and Universities*, Ron Dodge, Lance J. Hoffman, Daniel Ragsdale, and Tim Rosenberg, 2004).

COMPETITION OBJECTIVES

- Build a meaningful mechanism by which institutions of higher education may evaluate their programs;
- Provide an educational venue in which students are able to apply the theory and skills they have learned in their course work;
- Foster a spirit of teamwork, ethical behavior, and effective communication both within and across teams; and
- Open a dialog and awareness among participating institutions and students.

TERMINOLOGY

Throughout this document, the following terms are used:

- **Operations Team/Gold Team**: competition officials that organize, run, and manage the competition.
- **White Team**: competition officials that observe team performance in their competition area and evaluate team performance and rule compliance.
- **Red Team**: penetration testing professionals simulating external hackers attempting to gain unauthorized access to competition teams' systems.
- **Black Team**: competition support members that provide technical support, pick-up and deliver communications, and provide overall administrative support to the competition.
- **Blue Team/Competition Team**: the institution competitive teams consisting of students competing in a CCDC event.
- **Team Captain**: a student member of the Blue Team identified as the primary liaison between the Blue Team and the White Team.
- **Team Co-Captain**: a student member of the Blue Team identified as the secondary or backup liaison between the Blue Team and the White Team, should the Team Captain be unavailable (i.e. not in the competition room).
- **Team representatives**: a faculty or staff representative of the Blue Team's host institution responsible for serving as a liaison between competition officials and the Blue Team's institution.

COMPETITION RULES

1. **Competitor Eligibility**
    a. Competitors in CCDC events must be full-time students of the institution they are representing.
        i. Team members must qualify as full-time students as defined by the institution they are attending.
        ii. Individual competitors may participate in CCDC events for a maximum of five seasons. A CCDC season is defined as the period of time between the start of the first state event and the completion of the National CCDC event. Participation on a team in any CCDC event during a given season counts as participation for that entire season.
        iii. A competitor in their final semester prior to graduation is exempt from the full-time student requirement and may compete in CCDC events as a part-time student provided the competitor has a demonstrated record of full-time attendance for the previous semester or quarter.
        iv. If a team member competes in a qualifying, state, or regional CCDC event and graduates before the next CCDC event in the same season, that team member will be allowed to continue to compete at CCDC events during the same season should their team win and advance to the next round of competition.
    b. Competitors may only be a member of one team per CCDC season.

2. **Team Composition**
    a. Each team must submit a roster of up to 12 competitors to the competition director of the first CCDC event they participate in during a given CCDC competition season. Rosters must be submitted at least two weeks prior to the start of that event. All competitors on the roster must meet all stated eligibility requirements. No changes to the team roster will be permitted after the team competes in their first CCDC event. The competition team must be chosen from the submitted roster. A competition team is defined as the group of individuals competing in a CCDC event.
    b. Each competition team may consist of up to eight (8) members chosen from the submitted roster.
    c. Each competition team may have no more than two (2) graduate students as team members.
    d. If the member of a competition team advancing to a qualifying, state, regional, or national competition is unable to attend that competition, that team may substitute another student from the roster in their place prior to the start of that competition.
    e. Once a CCDC event has begun, substitutions or additions of team members are prohibited. A team must complete the competition with the team that started the competition.
    f. Teams or team members arriving after an event's official start time, for reasons beyond their control, may be allowed to join the competition provided a substitution has not already been made. Event coordinators will review the reason for tardiness and make the final determination.
    g. Each team will designate a Team Captain for the duration of the competition to act as the team liaison between the competition staff and the teams before and during the competition. In the event of the Team Captain's absence, teams must have an

identified team liaison serving as the captain in the competition space at all times during competition hours.

h. An institution is only allowed to compete one team in any CCDC event or season.

3. **Team Representatives**
    a. Each team must have at least one representative present at every CCDC event. The representative must be a faculty or staff member of the institution the team is representing.
    b. Once a CCDC event has started, representatives may not coach, assist, or advise their team until the completion of that event (including overnight hours for multi-day competitions).
    c. Representatives may not enter their team's competition space during any CCDC event.
    d. Representatives must not interfere with any other competing team.
    e. The representative, or any non-team member, must not discuss any aspect of the competition event, specifically event injections, configurations, operations, team performance or red team functions, with their team during CCDC competition hours and must not attempt to influence their team's performance in any way.

4. **Competition Conduct**
    a. Throughout the competition, Operations and White Team members will occasionally need access to a team's system(s) for scoring, troubleshooting, etc. Teams must immediately allow Operations and White Team members' access when requested.
    b. Teams must not connect any devices or peripherals to the competition network unless specifically authorized to do so by Operations or White Team members.
    c. Teams may not modify the hardware configurations of competition systems. Teams must not open the case of any server, printer, PC, monitor, KVM, router, switch, firewall, or any other piece of equipment used during the competition. All hardware related questions and issues should be referred to the White Team.
    d. Teams may not remove **<u>any</u>** item from the competition area unless specifically authorized to do so by Operations or White Team members including items brought into the team areas at the start of the competition.
    e. Team members are forbidden from entering or attempting to enter another team's competition workspace or room during CCDC events.
    f. Teams must compete without "outside assistance" from non-team members including team representatives from the start of the competition to the end of the competition (including overnight hours for multi-day events). All private communications (calls, emails, chat, texting, directed emails, forum postings, conversations, requests for assistance, etc.) with non-team members including team representatives that would help the team gain an unfair advantage are not allowed and are grounds for disqualification and/or a penalty assigned to the appropriate team.
    g. Printed reference materials (books, magazines, checklists) are permitted in competition areas and teams may bring printed reference materials to the competition.
    h. Team representatives, sponsors, and observers are not competitors and are prohibited from directly assisting any competitor through direct advice,

"suggestions", or hands-on assistance. Any team sponsor or observers found assisting a team will be asked to leave the competition area for the duration of the competition and/or a penalty will be assigned to the appropriate team.

i. Team members will not initiate any contact with members of the Red Team during the hours of live competition. Team members are free to talk to Red Team members during official competition events such as breakfasts, dinners, mixers, and receptions that occur outside of live competition hours.
j. Teams are free to examine their own systems but no offensive activity against other teams, the Operations Team, the White Team, or the Red Team will be tolerated. This includes port scans, unauthorized connection attempts, vulnerability scans, etc. Any team performing offensive activity against other teams, the Operations Team, the White Team, the Red Team, or any global asset will be immediately **disqualified** from the competition. If there are any questions or concerns during the competition about whether or not specific actions can be considered offensive in nature contact the Operations Team before performing those actions.
k. Teams are allowed to use active response mechanisms such as TCP resets when responding to suspicious/malicious activity. Any active mechanisms that interfere with the functionality of the scoring engine or manual scoring checks are exclusively the responsibility of the teams. Any firewall rule, IDS, IPS, or defensive action that interferes with the functionality of the scoring engine or manual scoring checks are exclusively the responsibility of the teams.
l. All team members will wear badges identifying team affiliation at all times during competition hours.
m. Only Operations Team/White Team members will be allowed in competition areas outside of competition hours.
n. Conspiracy of any kind will result in **immediate disqualification.**
    i. This includes conspiring with the Red Team (obtaining information, sharing/leaking information, etc.).
    ii. This includes conspiring with other Blue Teams (coordinated plans, obtaining information, sharing/leaking information, etc.).

5. **Internet Usage**
    a. Internet resources such as FAQs, how-to's, existing forums and responses, and company websites, are completely valid for competition use provided there is no fee required to access those resources and access to those resources has not been granted based on a previous membership, purchase, or fee. Only resources that could reasonably be available to all teams are permitted. For example, accessing Cisco resources through a CCO account would not be permitted but searching a public Cisco support forum would be permitted. Public sites such as Security Focus or Packetstorm are acceptable. Only public resources that every team could access if they chose to are permitted.
    b. Teams may not use any external, private electronic staging area or FTP site for patches, software, etc. during the competition. Teams are not allowed to access private Internet-accessible libraries, FTP sites, web sites, network storage, email accounts, or shared drives during the competition. All Internet resources used during the competition must be freely available to all other teams. The use of external collaboration and storage environments such as Google Docs/Drive is prohibited

unless the environment was provided by and is administered by competition officials. Accessing private staging areas or email accounts is grounds for disqualification and/or a penalty assigned to the appropriate team.

c. No peer to peer or distributed file sharing clients or servers are permitted on competition networks unless specifically authorized by the competition officials.
d. Internet activity, where allowed, will be monitored and any team member caught viewing inappropriate or unauthorized content will be subject to disqualification and/or a penalty assigned to the appropriate team. This includes direct contact with outside sources through AIM/chat/email or any other public or non-public services including sites such as Facebook. For the purposes of this competition inappropriate content includes pornography or explicit materials, pirated media files, sites containing key generators and pirated software, etc. If there are any questions or concerns during the competition about whether or not specific materials are unauthorized contact the White Team immediately.
e. All network activity that takes place on the competition network may be logged and subject to release. Competition officials are not responsible for the security of any information, including login credentials, which competitors place on the competition network.

6. **Permitted Materials**
    a. No memory sticks, flash drives, removable drives, CDROMs, electronic media, or other similar electronic devices are allowed in the room during the competition unless specifically authorized by the Operations or White Team in advance. Any violation of these rules will result in disqualification of the team member and/or a penalty assigned to the appropriate team.
    b. Teams may not bring any type of computer, laptop, tablet, PDA, cell phone, smart phone, or wireless device into the competition area unless specifically authorized by the Operations or White Team in advance. Any violation of these rules will result in disqualification of the team member and/or a penalty assigned to the appropriate team.
    c. Printed reference materials (books, magazines, checklists) are permitted in competition areas and teams may bring printed reference materials to the competition as specified by the competition officials.

7. **Professional Conduct**
    a. All participants, including competitors, coaches, White Team, Red Team, Ops Team, and Gold Team members, are expected to behave professionally at all times during all CCDC events including preparation meetings, receptions, mixers, banquets, competitions and so on.
    b. In addition to published CCDC rules, Host Site policies and rules apply throughout the competition and must be respected by all CCDC participants.
    c. All CCDC events are alcohol free events. No drinking is permitted at any time during competition hours.
    d. Activities such as swearing, consumption of alcohol or illegal drugs, disrespectful or unruly behavior, sexual harassment, improper physical contact, becoming argumentative, willful violence, or willful physical damage have no place at the competition and will not be tolerated.

e. Violations of the rules can be deemed unprofessional conduct if determined to be intentional or malicious by competition officials.
f. Competitors behaving in an unprofessional manner may receive a warning from the White Team, Gold Team, or Operations Team for their first offense. For egregious actions or for subsequent violations following a warning, competitors may have a penalty assessed against their team, be disqualified, and/or expelled from the competition site. Competitors expelled for unprofessional conduct will be banned from future CCDC competitions for a period of no less than 12 months from the date of their expulsion.
g. Individual(s), other than competitors, behaving in an unprofessional manner may be warned against such behavior by the White Team or asked to leave the competition entirely by the Competition Director, the Operations Team, or Gold Team.

8. **Questions, Disputes, and Disclosures**
    a. PRIOR TO THE COMPETITION: Team captains are encouraged to work with the Competition Director and their staff to resolve any questions regarding the rules of the competition or scoring methods before the competition begins.
    b. DURING THE COMPETITION: Protests by any team must be presented in writing by the Team Captain to the White Team as soon as possible. The competition officials will be the final arbitrators for any protests or questions arising before, during, or after the competition. Rulings by the competition officials are final. All competition results are official and final as of the Closing Ceremony.
    c. In the event of an individual disqualification, that team member must leave the competition area immediately upon notification of disqualification and must not re-enter the competition area at any time. Disqualified individuals are also ineligible for individual or team awards.
    d. In the event of a team disqualification, the entire team must leave the competition area immediately upon notice of disqualification and is ineligible for any individual or team award.
    e. All competition materials including injects, scoring sheets, and team-generated reports and documents must remain in the competition area. Only materials brought into the competition area by the student teams may be removed after the competition concludes.

9. **Red Team Attack Rules**
    a. Confine attack activity to the official target list located on the ScoreBot player page.
    b. No physical attacks without prior approval
    c. No physical contact with any other player or administrator
    d. No Distributed Denial of Service (DDoS) Attacks

MID-ATLANTIC CCDC COMPETITION SCENARIO

The country of Hackistan was recently liberated by a coalition of international forces led by the United States. On April 12 and 13, 2013, citizens of Hackistan will be voting in the country's first democratic election. Members of the US Military will be providing election oversight and security. Current intelligence reports indicate terrorist groups are planning on a series of physical and cyber-attacks to disrupt and/or control the election process.

Hackistan is divided into eight (8) voting districts. Each voting district will have their own self-contained electronic voting system and military support network including Command and Control (C2), classified and unclassified networks, etc.

An HQ detachment will be providing communication oversight and operating as a Network Operations Center (NOC)/Security Operations Center (SOC) for the event.

Each defending team represents a group of civilian contractors and is responsible for protecting their district's network assets, coordinating communications with upper echelons (HQ unit) and supporting deployed assets in the field.

E-VOTING SCENARIO

Spectators and players will receive a voter ID card when they register at the event. The ID card will be theirs to protect and have a fictional name and voting district (1-8, or maybe names) on it.

During the event, those with a voter ID card can approach the appropriate kiosk, swipe their card and place their vote. Voters may vote for:

- President and vice president
- Two district representatives
- New country name
- New country flag
- National animal

Straight party voting will be allowed.

RED CELL GOALS

Related to the Election

- Shutdown the voting system (lowest priority)
- Invalidate the election with greater than 100% of the registered voters to vote
- Invalidate the election with voters voting more than once
- Get their candidate elected (highest priority)

General Goals

- Obtain execute privilege on the defenders' systems
- Acquire flags
- Corrupt data
- Disable services

SCORING

General Information

All Blue Teams start with 0 points. Blue Teams are ranked against each other in order of highest (best) to lowest score.

Defending Teams will be scored across the following domains:

1. Service Functionality: All scored services must remain up, available and with a high degree of integrity. All services are given a predefined point value and will be checked periodically in Service Rounds. For each service that passes the necessary check the team will receive the appropriate number of points for that service. **The more service points a team receives, the better.**
2. Injects: During the course of the event, teams may receive any number of injects. An inject is any assigned task to be completed in the appropriate amount of time. Sample injects include creating policy documents, making technical changes and attending meetings. If the inject is completed on time and to standard, the team will receive the appropriate number of points. Team Captains may assign injects to specific team members for completion.
3. Preventing Red Team Activity: The Red Team (attackers) will have specific goals during this event. Sample goals include, compromising a server, stealing data or modifying records. Each Red Team goal is assigned a point value. If the goal is accomplished, the Red Team team/member is awarded the points and the Blue Team has a corresponding amount of points deducted from their score. For example; A Red Team player has the goal of obtaining a specific file off a Blue Team's mail server. The goal is worth 250 points. If the Red Team player acquires the file, they will receive 250 points and the victim Blue Team will have 250 points deducted from their score.
4. Blue Team Flags: The Blue Team will have flags to locate inside their *own* environment. The flags will come in two varieties: one will be informative and scenario-based; the other will be used to verify information integrity. A flag consists of two pieces of information. You'll be provided with one piece and must respond with the correct corresponding piece in order to get credit for locating the flag.

   Example Informational Flag:
   Provided: *What is the DNS resolvable hostname for the asset with an IP address of 192.168.1.100?*
   Answer: *flag.target1.com*

5. Supervisor Reporting: On Day Two of the competition, each Team Captain will meet twice, both times face-to-face, with the Chief Election Judge (CEJ) of Hackistan. More on how this is scored in the Supervisor Reporting section below.

Raw scores are used for these scoring metrics. However, at the end of each day's competition, Blue Teams are ranked using an ordinal scale, which is a measurement scale that assigns values to objects based on their ranking with respect to one another. For example, a first place finish at the end of day one in the Service Functionality scoring metric warrants an ordinal score of 1; a second place finish warrants an ordinal score of 2; up to an eighth place finish warranting an ordinal score of 8. This process is repeated for the Injects, Preventing Red Team Activity, Flags, and Supervisor Reporting scoring metrics.

The ordinal scores from the five (5) scoring metrics are then totaled for each Blue Team, yielding a combined ordinal score for each day of competition, which is used to rank the Blue Teams from first through eighth place. The winning Blue Team will be based on the lowest combined ordinal score obtained during the competition time.

Tie Breakers

In the event of a tie, the team with the highest raw inject score will win.

Service Rounds

The services are scored in Rounds. For each round, a Blue Team will be scored on:

1. Service Availability (are critical services available with a high degree of integrity?)
2. Data Confidentiality (has key data been stolen?)
3. System Compromises (have any systems been compromised by a Red Team?)

All systems are pinged before the service checks are run. If a system fails to respond to pings, it will be fined the total value of all services at Level 1 (see below). For example, a server has three services on it and each service is worth 1,000 points. If the server cannot be pinged, then the Blue Team will have 4,000 points subtracted from its total (1,000 points for the failed ping and 1,000 x 3 for each down service).

If the external interface of the firewall cannot be pinged, then no subsequent tests will be run and the maximum penalty of all services scored at Level 1 will be subtracted.

TCP is scored at three levels:

- **Level 1** is the TCP three-way handshake. If the port is not open, or does not complete the three-way handshake, then the team is fined the number of points for that service as listed in the Server/Services table (see above).
- **Level 2** is the service request. If Level 1 is passed (the port is open and the three-way handshake is completed), then the scoring engine makes a service request (e.g., HTTP get index.html). If the request is not processed, then the Blue Team is fined one-half the number of Level 1 points.
- **Level 3** is the integrity check. If Levels 1 and 2 are passed, then checks will be performed on the various flags to ensure their integrity has not changed since the start of the exercise. If the flag fails the integrity check, then the Blue Team is fined one fourth the number of Level 1 points.

UDP is scored at two levels:

- **Level 1** checks to see if the port is open by issuing a service request to that port. If the port is unresponsive, then the Blue Team will be fined the number of points for that service as listed in the Server/Services table (see above).
- **Level 2** is the integrity check. If Level 1 is passed, then checks will be performed on the various flags to ensure their integrity has not changed since the start of the exercise. If the flag fails the integrity check, then the Blue Team is fined one half the number of Level 1 points.

Dependent Services

The services that are being scored represent critical services. The scored services may rely on NON-SCORED services. If a service is NOT being scored it does NOT mean that it is NOT necessary for a scored service to run. Shutting down a dependent service may result in points being subtracted from your team's service score.

Non-Service Scored Assets

There may be non-service scored assets within each Blue Team network. A non-service scored asset is one that is on the official Services List and NAT'd to the outside.

As the name of these assets implies, these assets are not checked for ICMP or any network services. Any service outage or lack of network connectivity on these assets will NOT impact the service portion of a team's score.

Service Data Confidentiality

The Scoring Engine will be checking for stolen flags. A flag could be a file in a file system or an entry in a database table, or even a username/password combination. If a flag is successfully stolen from a Blue Team and submitted for scoring, points will be deducted to the Blue Team score.

Red Team

The Red Team will be attempting to find and exploit weaknesses in each Blue Team's environment. All systems reachable by the Red Team within the Blue Team's internal and external IP address space are valid targets. This includes systems that are not being scored.

The Red Team will be using Phone Home scripts to prove execute privileges on a scored asset. The Phone Home script connects to the Scoring Server across the game network. Any attempts to block the Phone Home script by using source or destination IP addressing is forbidden, unless the Operations Team gives permission (permission would be given after a successful Incident Response report is filed and reviewed by the attending Law Enforcement representative – see the Scoring: Law Enforcement section below).

Any Phone Home from a hidden asset does NOT impact the Blue Team's score. A Phone Home from a hidden asset DOES count towards the score of a Red Team member. As a result, the total number of Red Team Phone Homes may be more than the combined number of Phone Homes from all Blue Teams.

Since each Blue Team has access to their assets through the vSphere client, any Phone Home from a non-service scored asset DOES impact the Blue Team's overall score, with the value of the Phone Home being added to the Blue Team's score.

Blue Teams have points deducted from their score in the event of a successful Phone Home, per round. Teams may obtain a portion of their points back based on conducting an investigation. If the officer running the investigation thinks they have enough evidence to obtain a search warrant, points may be returned to the Blue Team. If the officer running the investigation thinks they have enough evidence to obtain an arrest warrant, additional points will be returned to the Blue Team.

Injects

Throughout the competition, the Blue Team will be presented with injects. The injects will vary in nature and be weighted based upon the difficulty and time sensitivity of the tasking. Tasks may contain multiple parts. Some examples: setting up an Intranet web server; performing a zone transfer between a secondary DNS server and the primary DNS server.

Injects can be delivered through any number of methods and will include the point values and time restrictions associated with the task. Upon completion of the injects, the performance of the Blue Team will be noted and scores adjusted.

Supervisor Reporting

On Day Two of the competition, each Team Captain will meet face-to-face with the Chief Election Judge (CEJ) of Hackistan.

During the initial meeting (five minutes, timed), the CEJ will expect to be briefed on the current status of the elections process, number of users impacted by downed systems, as well other items the Team Captain will consider relevant for the CEJ to know. At this initial meeting, each Team Captain will be given action items to complete and a fixed time (4 hours). Items may include a written status report, a high-level remediation plan, a resources inventory, a request for prioritized additional resources subject to budget constraints and other similar management factors.

During the second session (ten minutes, timed), each Team Captain will meet with the CEJ and will have a chance to present written and verbal responses from the first meeting and provide updates on any changes that transpired.

Each team will be scored using the following metrics:

- Oral presentation skills
- Writing skills
- Clarity of communicating the situation
- Ability to rise above techno-babble
- Creativity in reacting to new information

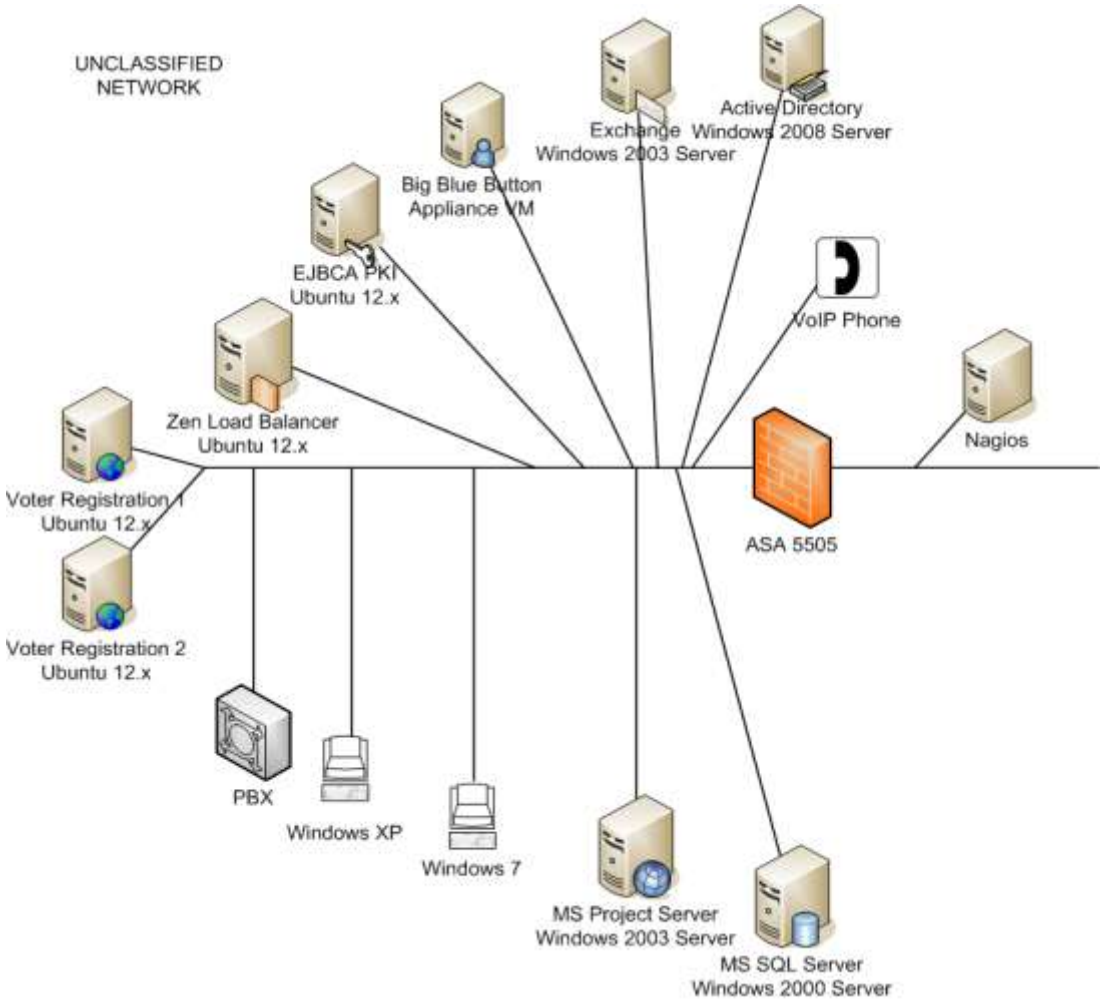
Like the other scoring metrics, an ordinal scale will be used to rank the teams.

Incident Reporting

All Blue Teams must complete at least two Incident Response forms and open two cases with the attendant Law Enforcement Agents in attendance. Incident Response forms will be provided.

DEFENDER'S NETWORK (UNCLASSIFIED)

Each Blue Team will be responsible for defending the following assets:



IP ADDRESS TABLE

All teams have the same internal IP address space of 172.16.1.x

Team	External IP Space
Anne Arundel	192.168.22.x
Capitol College	192.168.23.x
Craven Community College	192.168.24.x
Millersville University	192.168.25.x
North Carolina State University	192.168.26.x
Radford University	192.168.27.x
University of Maryland, College Park	192.168.28.x
University of Maryland, University College	192.168.29.x
HQ	192.168.21.x

SYSTEM INFORMATION

Firewall

The firewall is setup in a two-segment configuration (DMZ and outside). Network Address Translation (NAT) and traffic flow is very important.

DMZ Segment

This is a list of your systems and SOME (not ALL) of the services running on them. A final list of services will be provided at the event:

- **Microsoft Active Directory (AD) Server** - Windows Server 2008: This is your primary domain controller and is responsible for managing user and system information across the enterprise. There are several hundred users. The AD server is also your primary DNS server. As your primary DNS server, it must allow zone transfers and other inbound DNS requests. Failure to resolve DNS lookups from outside your network could result in service problems/penalties.
- **Microsoft Exchange Server** – Windows Server 2003: This server provides all email communication for the team. This includes all necessary email services such as IMAP, SMTP, POP3 and Outlook Web mail. This server also functions as your team's secondary DNS and Active Directory Server.
- **Big Blue Button** – Appliance Version: BigBlueButton is an open source web conferencing system built on over fourteen open source components to create an integrated solution that runs on Mac, UNIX, and PC computers. In the true sense of open source, we invite you to try out and participate in our project.
- **EJBCA PKI** – Ubuntu 12.x: EJBCA is an enterprise class PKI Certificate Authority built on JEE technology. It is a robust, high performance, platform independent, flexible, and component based CA to be used stand-alone or integrated in other JEE applications.
- **MS Office Project Server 2003** – Windows Server 2003: Microsoft's Project Management server.
- **MS SQL Server** – Windows Server 2000: MS SQL Server to support the Office Project Server.
- **Voter Registration (1 and 2)** – Ubuntu 12.x: A Ruby on Rails Application that generates a National Voter Registration Form PDF from a web form. Includes the guidelines for the National Voter Registration form in all states.
- **Zen Load Balancer** – Ubuntu 12.x: Zen Load Balancer is a complete solution for load balancing to provide a high availability for TCP and UDP services and data line communications, targeted to become a professional open source product in networking for distributed systems. This will be used to balance the load between the two Voter Registration Servers
- **FreePBX Server** – Appliance: FreePBX is an easy to use GUI (graphical user interface) that controls and manages Asterisk, the world's most popular open source telephony engine software. FreePBX has been developed and hardened by thousands of volunteers over tens of thousands man hours. FreePBX has been downloaded over 5,000,000 times and estimates over 500,000 active phone systems. If you don't know about FreePBX, you are probably paying too much for your phone system.

Outside Segment

Nagios and Cacti – Fedora Core 7: This server is physically located outside your firewall, but within your team's public IP range. This machine is running two services:

- **Nagios**: the Nagios server has its configuration files located in /etc/nagios3. The team configuration is located in /etc/nagios3/objects/team<#>.cfg. Each of the competition servers has been preconfigured to monitor the same services that are being scored.
- **Cacti**: is a complete network graphing solution designed to harness the power of RRDTool's data storage and graphing functionality. Cacti provides a fast poller, advanced graph templating, multiple data acquisition methods, and user management features out of the box. All of this is wrapped in an intuitive, easy-to-use interface that makes sense for LAN-sized installations up to complex networks with hundreds of devices. The Cacti server is using SNMP to monitor your firewall's CPU, RAM and throughput.

Hidden Asset

There may be hidden assets within each Blue Team's network. A Hidden Asset is one that has access to, or is directly on the Blue Team's internal network; however, you do not have access to the server through the vSphere client, nor is the asset on your official Services List.

This document is proprietary and confidential and intended solely for the use of the National CyberWatch Mid-Atlantic Collegiate Cyber Defense Competition participants. Unauthorized duplication and/or distribution in whole or in part is expressly prohibited. Portions of this document are covered under the following copyrights:

© 2013 iSIGHT Partners
© 2013 The University of Texas at San Antonio
© 2013 Prince George's Community College
All Rights Reserved